Cloud Architecture

Oracle APEX: CRUD operations

Summary Notes

Contents

Lesson objectives

By the end of this lesson, you should be able to:

- Discuss Oracle APEX cloud architecture
- Explore SQL commands in APEX
- Perform SQL READ operations

Lesson prerequisites

Basic knowledge of CRUD operations

CRUD stands for Create, Read, Update, and Delete, which are the four basic operations performed on data in a database. These operations form the basis of managing data in a database and are used to create, retrieve, update, and delete records in a database table. Here's a brief description of each operation:

- **Create**: This operation is used to insert new records into a database table.
- **Read**: This operation is used to retrieve data from a database table.
- **Update**: This operation is used to modify existing records in a database table.
- **Delete**: This operation is used to delete records from a database table.

These operations are typically performed using SQL statements such as INSERT, SELECT, UPDATE, and DELETE, respectively. The CRUD operations are used to perform the majority of database operations and play a crucial role in database management systems.

Familiar with DML statements

DML stands for Data Manipulation Language, and refers to a set of SQL commands used to manage and manipulate data stored in a relational database. The main DML statements are:

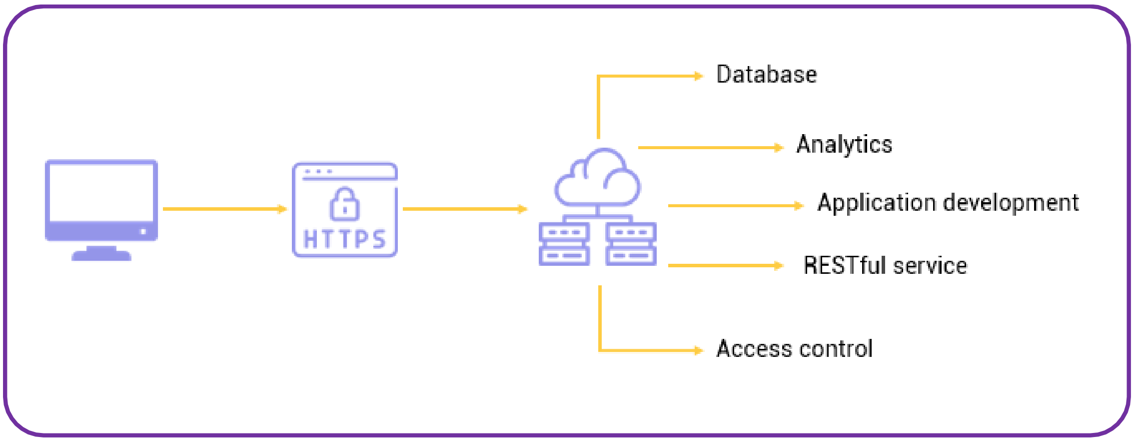
- **SELECT** - used to retrieve data from a database table.
- **INSERT** - used to insert new data into a table.
- **UPDATE** - used to modify existing data in a table.
- **DELETE** - used to delete data from a table.

These statements allow users to perform common data manipulation tasks, such as adding new records, updating existing ones, and retrieving specific data from a database. They are essential for managing and maintaining data within a database, and form a fundamental part of SQL.

Oracle APEX cloud architecture

The Oracle APEX cloud architecture various from traditional architecture in that a local client only requires an internet connection to access the cloud resources. The client connects to the internet using the HTTPS protocol, which in turn provides access to an array of services hosted by Oracle APEX. These services include:

- Databases
- Analytics
- Application development
- RESTful services
- Access control



Practical demonstration - Source code

```
--Select all customers
select *
from customers

--select customer name and website
select name, website
from customers

--select name and address of all customers in NY
select name, address
from customers
where substr(address,-2) = 'NY'

--select product name and category id
select product_name,category_id
from products
where category_id = 5

--select the product name, std cost and list price of all products in category 5
select product_name, standard_cost, list_price
from products
where category_id = 5
```

```sql
--select the product name, std cost and list price of all products in category 5 join
cat name
select product_name, standard_cost, list_price, category_name
from products join product_categories
using (category_id)

--select the product name, std cost and list price of all products in category 5 join
name and calculate profit
select product_name, standard_cost, list_price, category_name, list_price-standard_cost
as profit
from products join product_categories
using (category_id)

--same as above where cat id is 5
select product_name, standard_cost, list_price, category_name, list_price-standard_cost
as profit
from products join product_categories
using (category_id)
where category_id = 5
```